Presentation of a paper:

Modeling the Choice of Residential Location
Daniel McFadden (1978)

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Contents

Background of the authors

Academic

Daniel McFadden

Nobel laureate, attended University of Minnesota B.S. in Physics , later Ph.D Behavioral Science (Economics). He has been a faculty member of University of Berkeley, MIT & University of Southern California (emeritus).

Choice of Residential Location

The whole paper starts with a key classical assumption that the consumers are rational, utility-maximizing consumers.

$$P_{cn} = Prob[U_{cn} > U_{bm} \quad | \quad _{bm} \neq _{cn}]$$

- U_{cn} - utility of choosing property in community C
- U_{bm} - utility of choosing property in community B

Using econometrics U_{cn} & U_{bm} would be estimated with a error term, which translated leads us to the following probability equation:

$$P_{cn} = \int_{\varepsilon_{cn}=-\infty}^{+\infty} F_{cn}[(V_{cn}) + \varepsilon_{cn} - V_{dm}]d\varepsilon_{cn}$$

Multinomial Logit

Using the previous definitions of the probability using Econometric estimation we can express the probability in a classification form as follows:

$$P_{cn} = \frac{exp(V_{cn})}{\sum_{b=1}^{C} \sum_{m=t}^{N_b} exp(V_{bm})}$$

And for the framework we use in Residential location we use the common benefits to model the estimate utility as a vector of independent variables X_{cn} which are community specific and Y_c which is property specific.

$$V_{cn} = \beta \cdot X_{cn} + \beta_2 \cdot Y_c$$

Nested Logit

Using the previous definitions of the probability using Econometric estimation we can express the probability in a classification form as follows:

$$P_{cn} = \frac{\exp\left(\beta_2 \cdot Y_c + (1-\beta_2) \cdot I_c\right)}{\sum_{b=1}^{C} \exp\left(\beta_2 \cdot Y_b + (1-\beta_2) \cdot I_b\right)}$$

Generalized Model

Defining a general

$$\bar{U} = \int_{\varepsilon=-\infty}^{+\infty} max(V_i + \varepsilon_i) \cdot F(e) \cdot d\varepsilon_{cn}$$

Which is linearized could be should as:

$$\bar{U} = logG[exp(V_1), \ldots, exp(V_j)] + e(0.57721)$$

From which we could define:

$$P_i = \frac{\delta \bar{U}}{\delta(V_i)}$$

Final Multinomial Logit

$$P_c = \frac{\exp\left(\frac{\beta_2 Y_c + \beta X_c^* + (1-\beta_2)\log r_c + \frac{\omega_c^2}{2}}{1-\sigma}\right)}{\sum_{b=1}^{C} \exp\left(\frac{\beta_2 Y_b + \beta X_b^* + (1-\beta_2)\log r_b + \frac{\omega_b^2}{2}}{1-\sigma}\right)}$$

- $\beta_2 Y_j$: contribution of observable location attributes Y.
- βX_j^*: household–location interactions or other covariates.
- $(1-\beta_2)\log r_j$: (dis)utility from log rent.
- $\frac{\omega_j^2}{2}$: variance correction from a normal random component ω_j.
- Division by $1-\sigma$ scales sensitivity to utility differences (require $\sigma < 1$).

Ending remarks

I am delighted for your attention.

Q & A?